

SOEM 1.3.1 vs 2.0.0

Version differences and license implications for the EtherCAT master project — Xilinx Zynq UltraScale+ MPSoC / PREEMPT-RT Linux. Generated 2026-08-31.

1. License change (the decisive difference)

The license changed between the two versions in a way that directly affects whether this master application can stay proprietary.

SOEM 1.3.1 — GPLv2 with a linking exception

The source header shipped in the workspace states:

“As a special exception, if other files instantiate templates or use macros or inline functions from this file, or you compile this file and link it with other works to produce a work based on this file, this file does not by itself cause the resulting work to be covered by the GNU General Public License. However the source code for this file must still be made available in accordance with section (3) of the GNU General Public License.”

Practical effect: you may keep the master application (e_lmo_com.c, ecat_diag.c, ecat_foe.c, the config GUI, etc.) **closed / proprietary**. The only copyleft obligation is that if you *modify SOEM's own source files*, those modified files must be made available. Merely linking your app against SOEM does not force your app open.

SOEM 2.0.0 (and current master) — GPLv3 or commercial, no exception

The 2.0.0 LICENSE.md states:

“This software is dual-licensed. GPL version 3 — distributed under GPLv3; you are allowed to use this software for an open-source project with a compatible license. Commercial license — also available with options for support and maintenance (sales@rt-labs.com). If you intend to use this stack in a commercial product, you likely need to buy a license.”

The linking exception is **gone**. Under plain GPLv3, statically linking your master into a product makes the **combined binary a GPLv3 work** — you must offer the full corresponding source of your application under GPLv3, or obtain rt-labs' commercial license.

License comparison

	SOEM 1.3.1	SOEM 2.0.0
Base license	GPLv2 + linking exception	GPLv3, or commercial
Link a closed app to it?	Yes — app stays proprietary	No — GPLv3 copyleft covers the whole combined work
Obligation when you ship	Publish only modified SOEM files	Publish the entire linked program under GPLv3, or buy a commercial license
Patent / anti-tivoization	GPLv2 terms	GPLv3: stronger patent grant + anti-tivoization (Installation Information)

Embedded-target note: GPLv3's anti-tivoization clause is relevant because this runs on a locked-down Zynq/embedded board. Distributing GPLv3 code on hardware you lock down can trigger obligations to provide “Installation Information” so the user can run modified versions.

License bottom line

- Internal / research use, or you are happy to release your code under GPLv3: the upgrade is fine.
 - Closed commercial product: 1.3.1→2.0.0 flips you from “link freely, keep code closed” to “open your whole app **or** pay rt-labs.” That cost often outweighs the technical gains.
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2. Technical differences (what you gain by upgrading)

Bandwidth / performance

- **Overlapping IOMap** (`ec_config_overlap_map`): RxPDO and TxPDO share the same logical address region, roughly halving process-data frame size. 1.3.1 has no equivalent. The main concrete runtime win — matters if you scale slave count or push the cycle rate (your 250 μ s / 4 kHz target).

Robustness (8+ years of fixes)

- Mailbox handling — including the emergency-reread behavior worked around in the simulator — plus EEPROM/SII, DC delay calculation, LRW/segmented frames, and FoE/SoE/EoE fixes.
- 1.3.1 (~2013/2015) is effectively end-of-life: no upstream fixes, and modern GCC flags much of its code.
- Redundancy: `ec_init_redundant` path has refinements in 2.0.0.

API / architecture

- 2.0.0 folds all sub-arrays (`port`, `slavelist`, `grouplist`, `esibuf`, `elist`, `idxstack`) into a single `ecx_context_t`. In 1.3.1 a context means wiring ~a dozen separate globals. Enables multiple masters per process and is more thread-safe (relevant since the recovery thread and RT loop both touch the bus).
- **CMake** build with proper toolchain files instead of hand-rolled per-arch Makefiles — cleaner aarch64 cross-compile and out-of-tree builds.
- The global `ec_*` API remains available via `EC_VER1`, so most of the app port is mechanical.

Protocol coverage you currently skip

- **CoE Complete Access** and more robust SDO-info (OD list/description) — exactly why an online object-dictionary browser was deferred; far more usable in 2.0.0.
- More complete **EoE** (diagnostics over EtherCAT) and **SoE/IDN** handling for non-CoE drives.

What you would NOT gain

- Still a soft-RT, single-process, non-hard-DC master. No fundamental architecture change.
 - Everything already built — `ecat_diag` (0x0300 block), EMCY via `ec_iserror/ec_elist2string`, identity via `ec_slave[] . eep_*`, topology via `ec_slave[] . topology`, DC 0x092C, FoE via `ec_F0Ewrite` — exists in 2.0.0 under the same `EC_VER1` names.
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3. Migration cost, specific to this repository

- Rebuild the aarch64 static libs (`Libsoem/Liboshw/Libosal`) for Zynq via a CMake toolchain file — the project currently ships prebuilt .a files.
- Port the no-hardware sim: `nicdrv_sim.c` replaces the oshw `nicdrv`, and 2.0.0's port interface (`ecx_port_t`) is context-based with changed signatures — the shim and the sim Makefile need reworking.

- Optionally keep the global `ec_*` wrappers (`EC_VER1`) to minimize app churn, then re-run the full test suite.
 - Offline PC has no internet: fetch the 2.0.0 source on the build machine and copy it in.
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4. Recommendation

- If the current bus (2–3 Elmo CSP axes, single master) meets timing and the custom parity features are tested and working, staying on 1.3.1 is low-risk and functionally fine.
- Upgrade mainly buys overlapping-IOMap bandwidth, accumulated bug fixes/robustness, OD-browsing / EoE, and a supported baseline — weigh against the `sim/nicdrv` + `CMake` + cross-lib re-port effort.
- **License is often the deciding factor:** if this may become a closed commercial product, the GPLv2+exception → GPLv3/commercial change is the single biggest reason to stay on 1.3.1 (or budget for an rt-labs commercial license).

This document is engineering guidance, not legal advice. If a commercial product is in scope, confirm license obligations with counsel or request a quote from rt-labs (sales@rt-labs.com) before committing to SOEM 2.0.0.